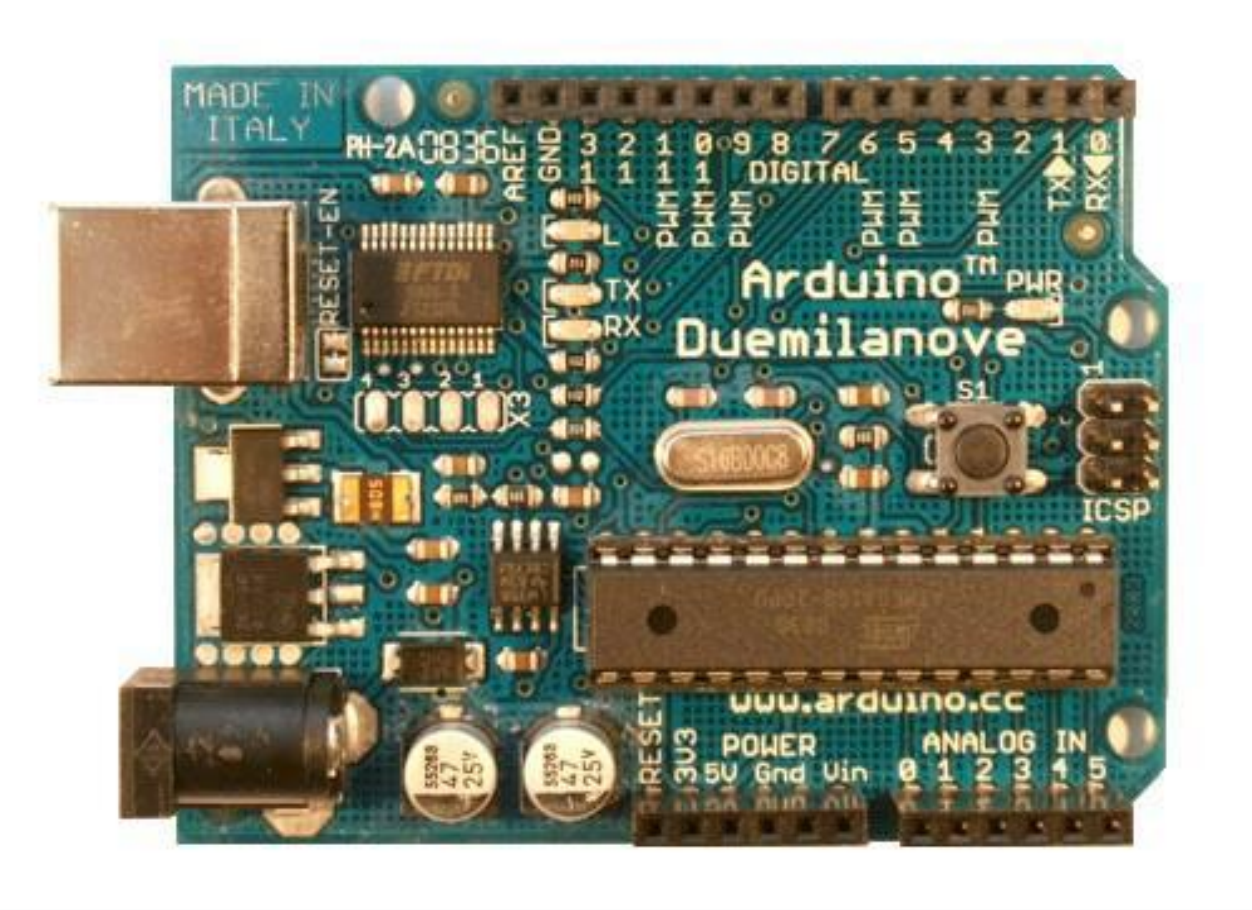
ARHITEKTURA RAČUNALA

Građa mikroupravljača ATMEGA328P

Građa mikroupravljača ATMEGA328p

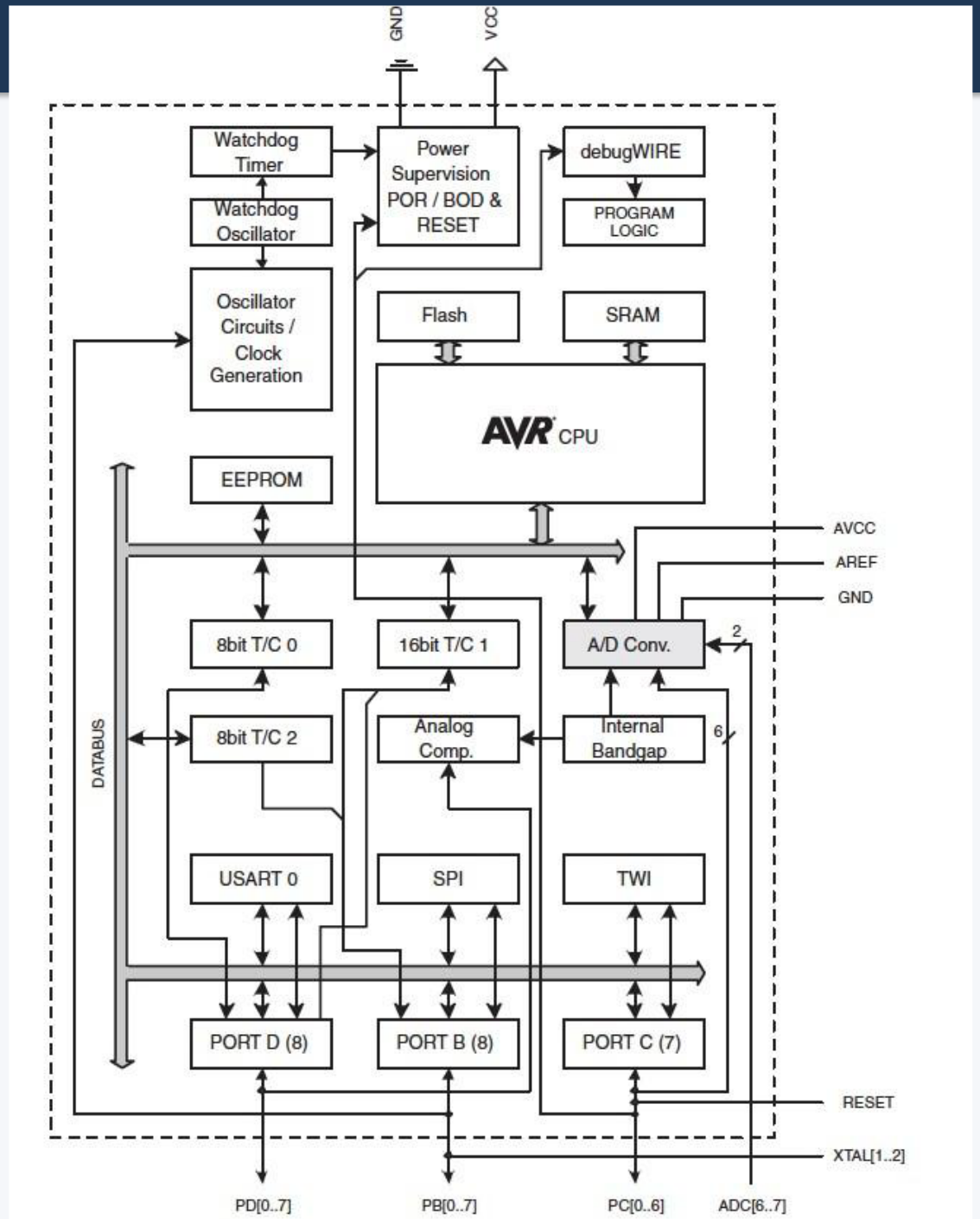
Atmega 328p

- arhitektura and instruction set
- I/O pins
- Arduino C++ language



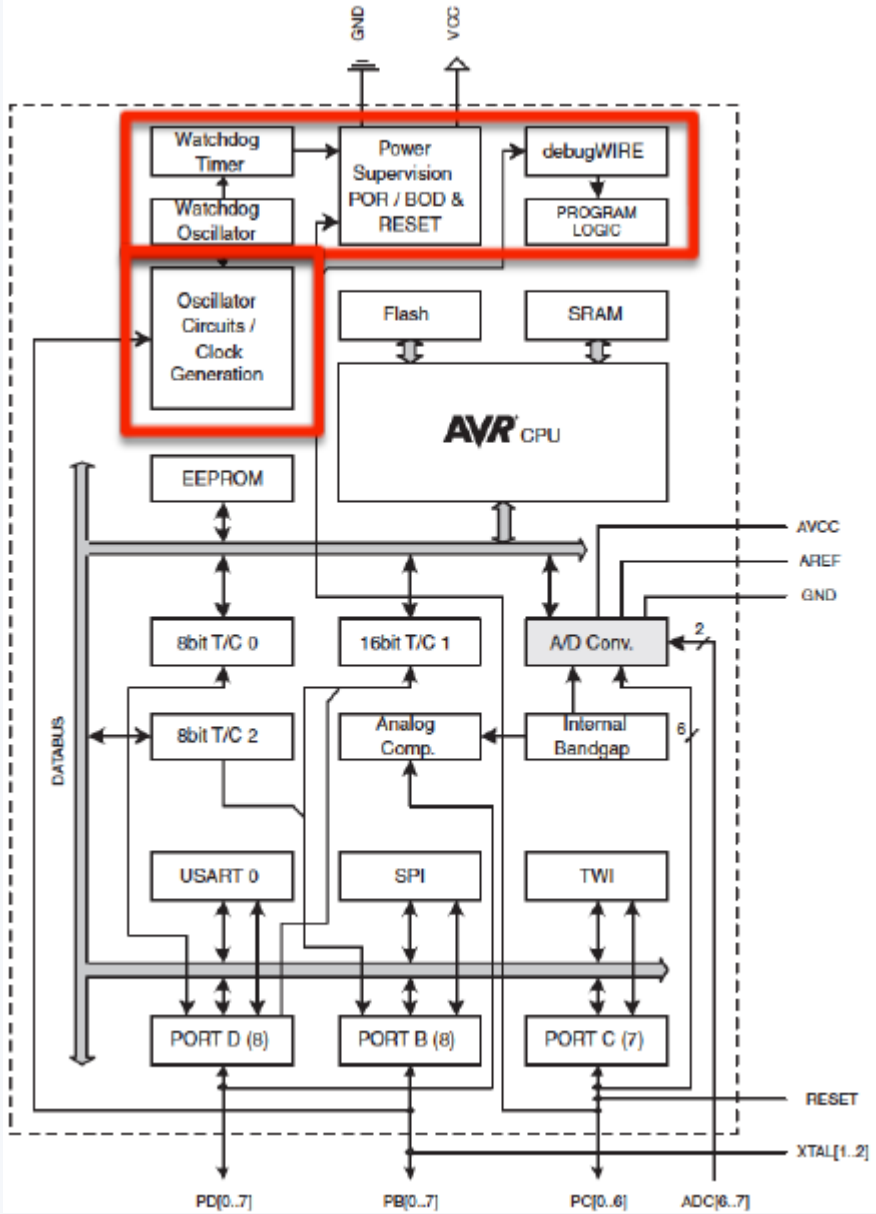
Adresa memorije

- Arhitektura



Arhitektura ATMEGA 328p

- Napajanje
- Oscilator
- Watchdog
- Programska logika



Arhitektura ATMEGA 328p

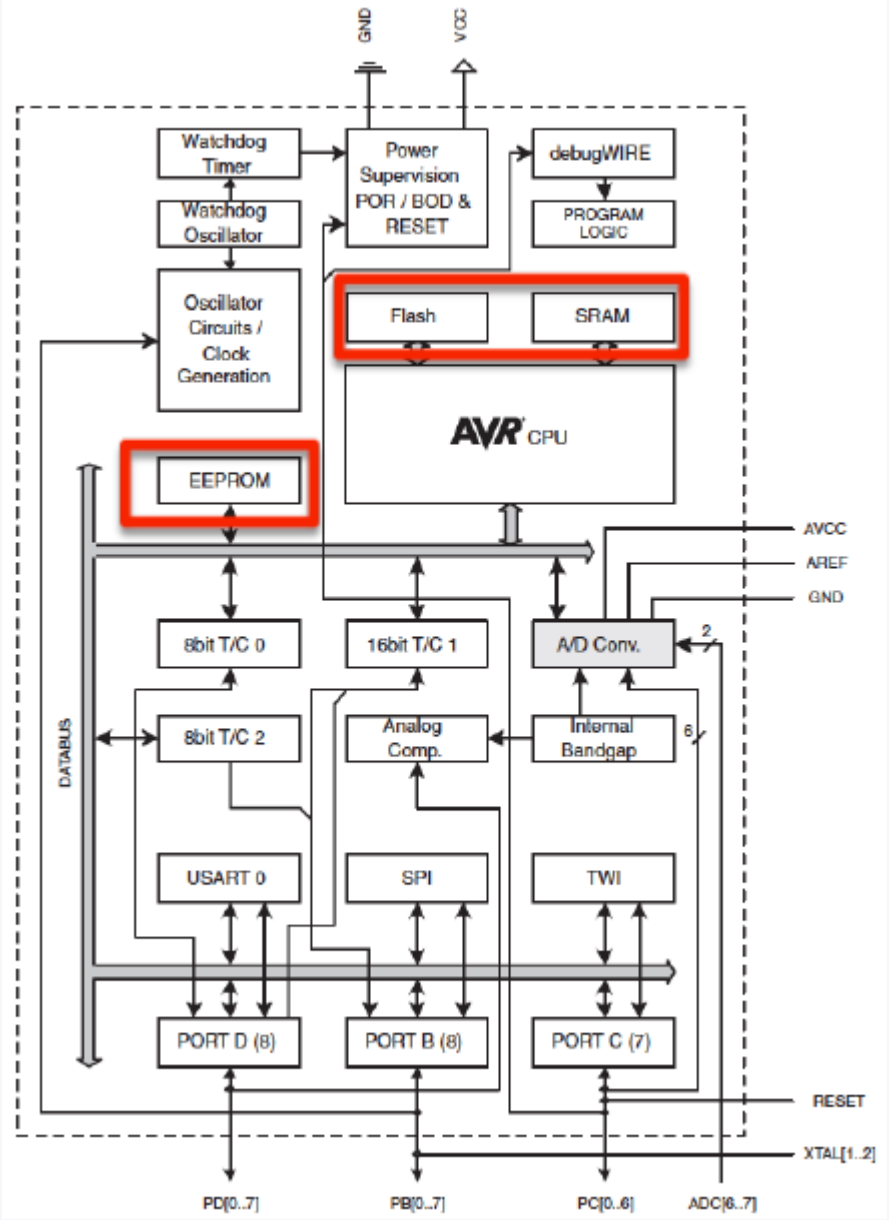
Harvard arhitektura

Flash (32K) (15-bit adrese)

- Programska memorija – read only
- memorija koja ne gubi podatke isključenjem napajanja
- Pohranjuje podatke koristeći naredbu PROGMEM keyword

SRAM (2K)

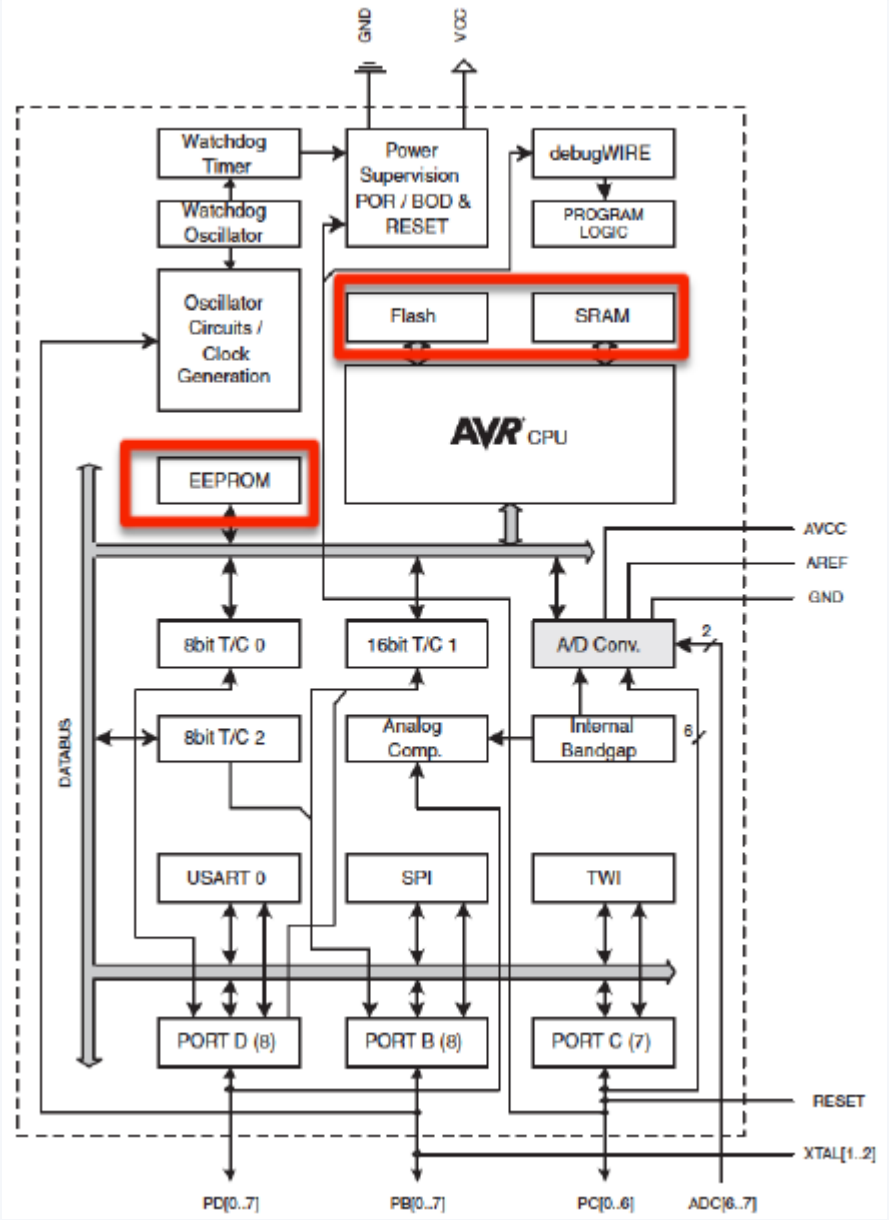
- Trenutne vrijednosti
- briše se gašenjem napajanja
- Ograničeni prostor



Arhitektura ATMEGA 328p

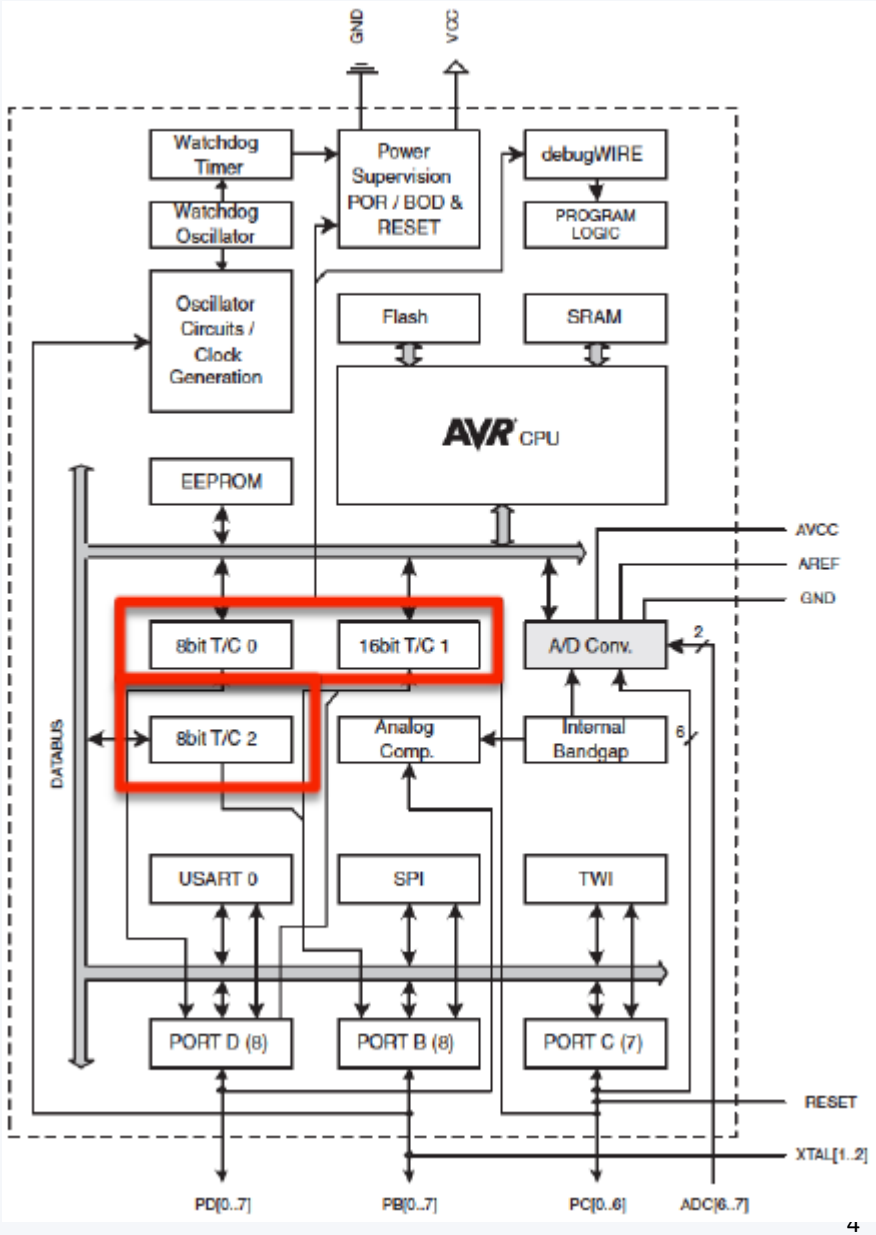
EEPROM

- Long-term data – sadrži podatke koje je potrebno na dulje vrijeme pohraniti
- EEPROM library → biblioteka koja se može koristiti za pohranu i ispis te memorije



Arhitektura ATMEGA 328p

- 3 vremensko-brojačka sustava
  - Jednostavno postavljanje
  - Moguć odabir
    - Frekvencije
    - Vrijednosti brojača
  - Mogućnost generiranja prekida
  - Mogućnost generiranja PWM signala (8 bitne vrijednosti)



Arhitektura ATMEGA 328p

PROGRAMSKA IZVEDBA

delay(*ms*)

Čeka milisekundu prije nego nastavi s programom

delayMicroseconds(*us*)

Čeka mikrosekundu prije nego nastavi s programom

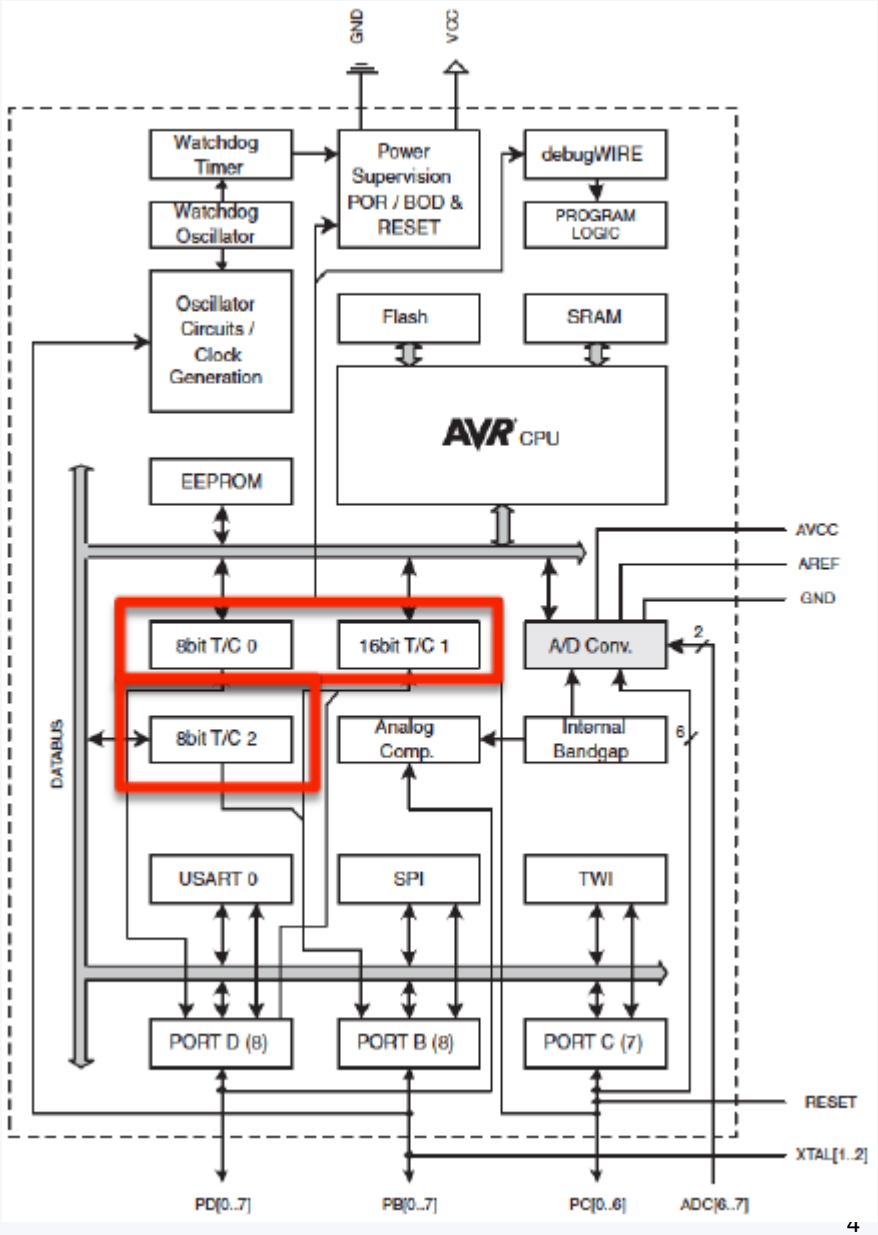
unsigned long millis( )

Vraća vrijednost u obliku milisekundi od kad je program krenuo

unsigned long micros( )

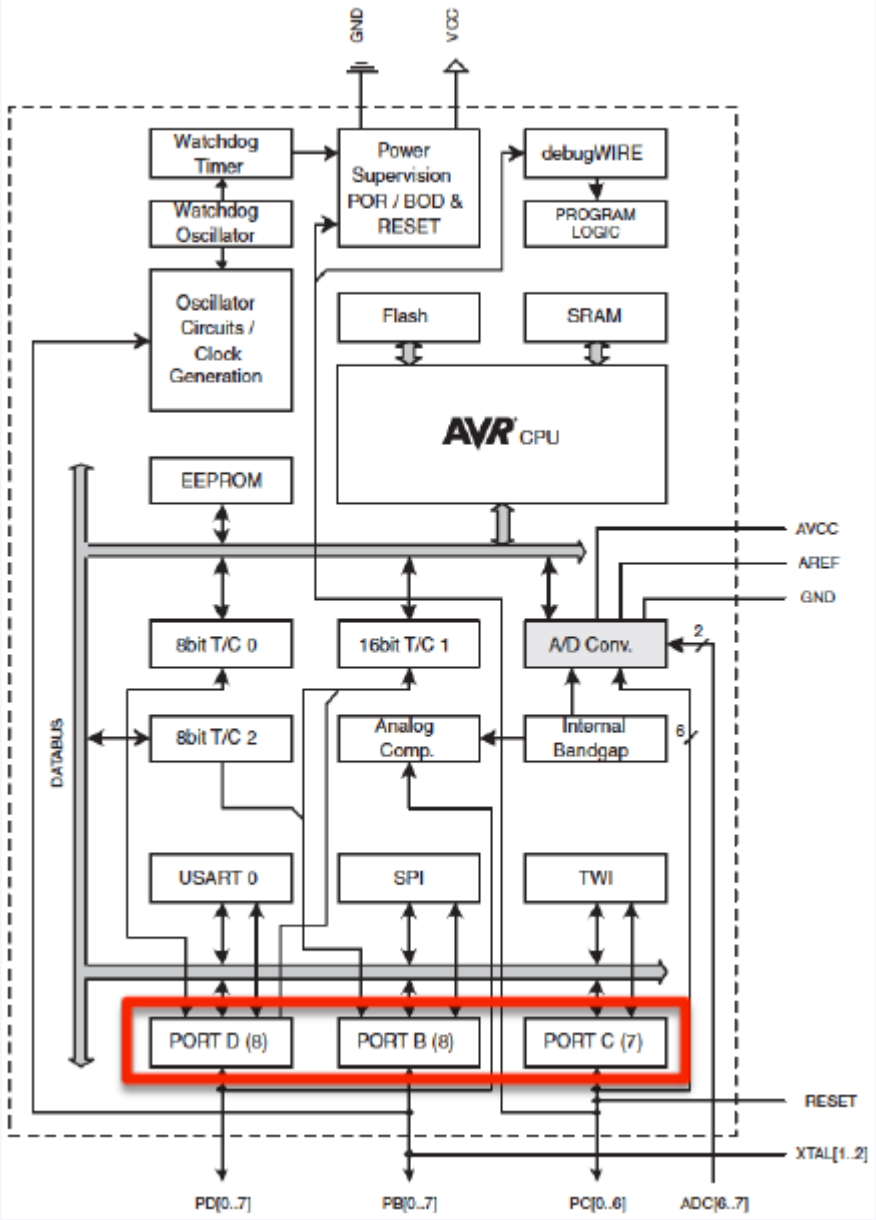
Vraća vrijednost u obliku mikrosekundi od kad je program krenuo

Rezolucija 4 mikrosekunde



Arhitektura ATMEGA 328p

- Sučelje za pinove
  - Svaki pin direktno programirljiv
  - Potrebno definirati
    - Smjer
    - Vrijednost
    - Pull_up
  - Vrste pinova
    - Analog vs. Digital
    - Clocks
    - Reset



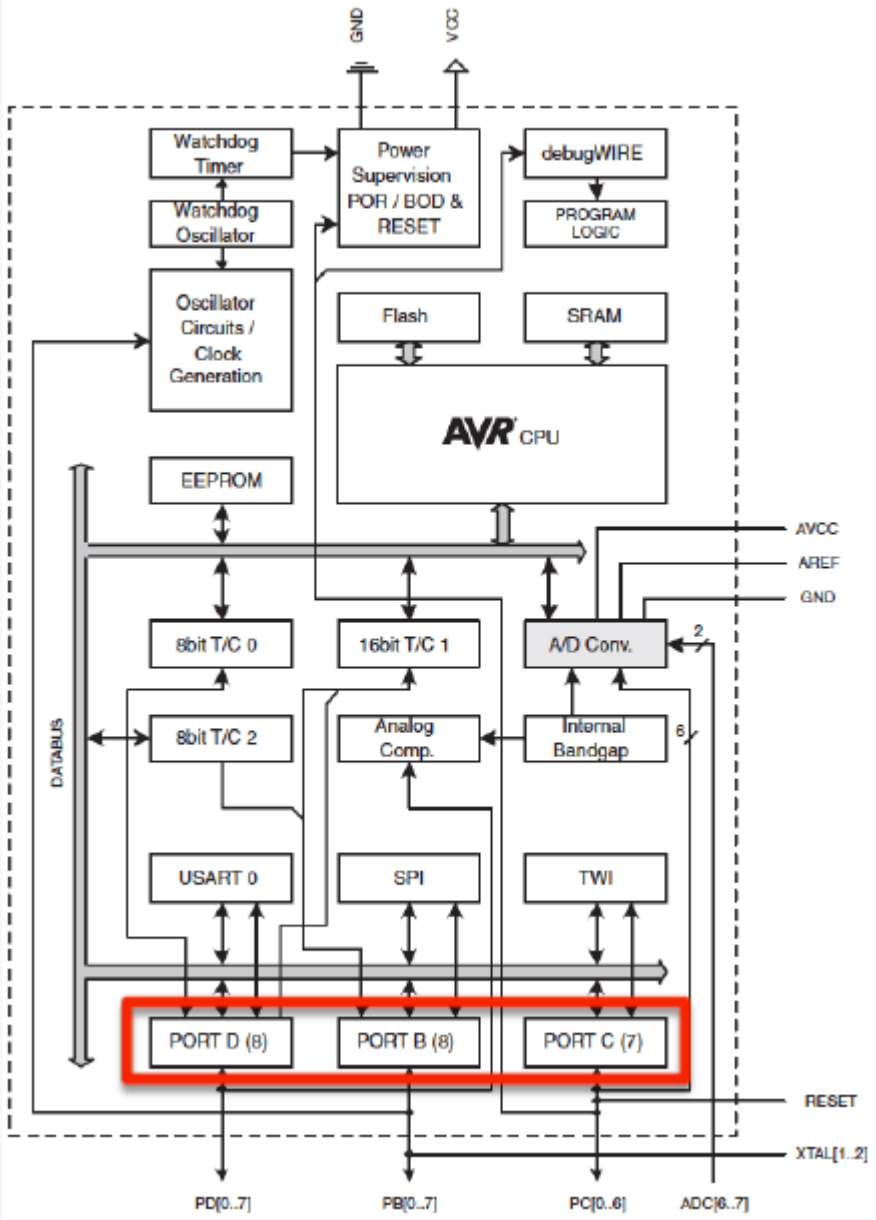
Arhitektura ATMEGA 328p

- **I/O Ports**
- ***3 8-bit Ports (B, C, D)***
- ***Each port controlled by 3 8-bit registers*** → Each bit controls one I/O pin

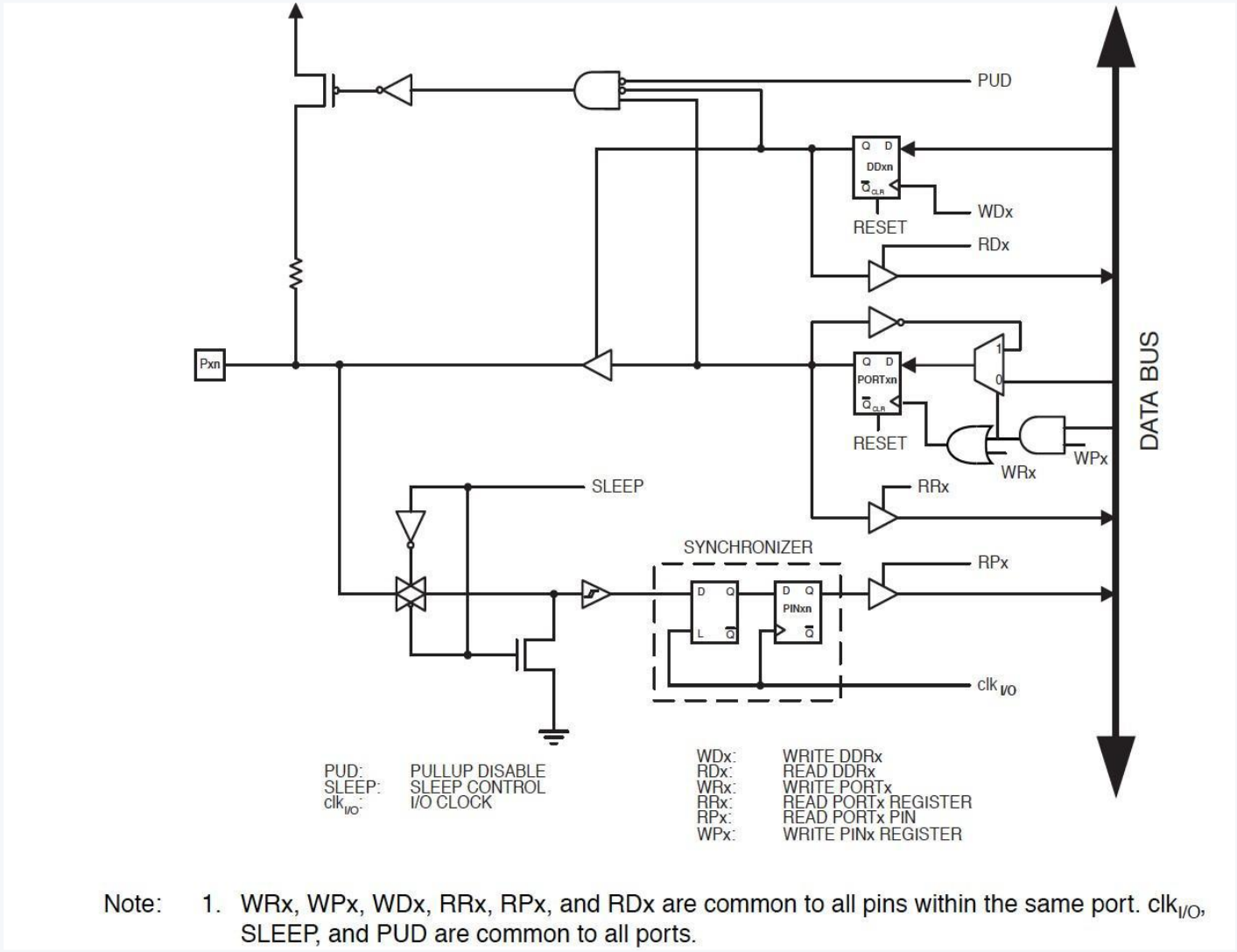
DDRx – Direction register Defines whether a pin is an input (0) or and output (1)

PINx – Pin input value Reading this “register” returns value of pin

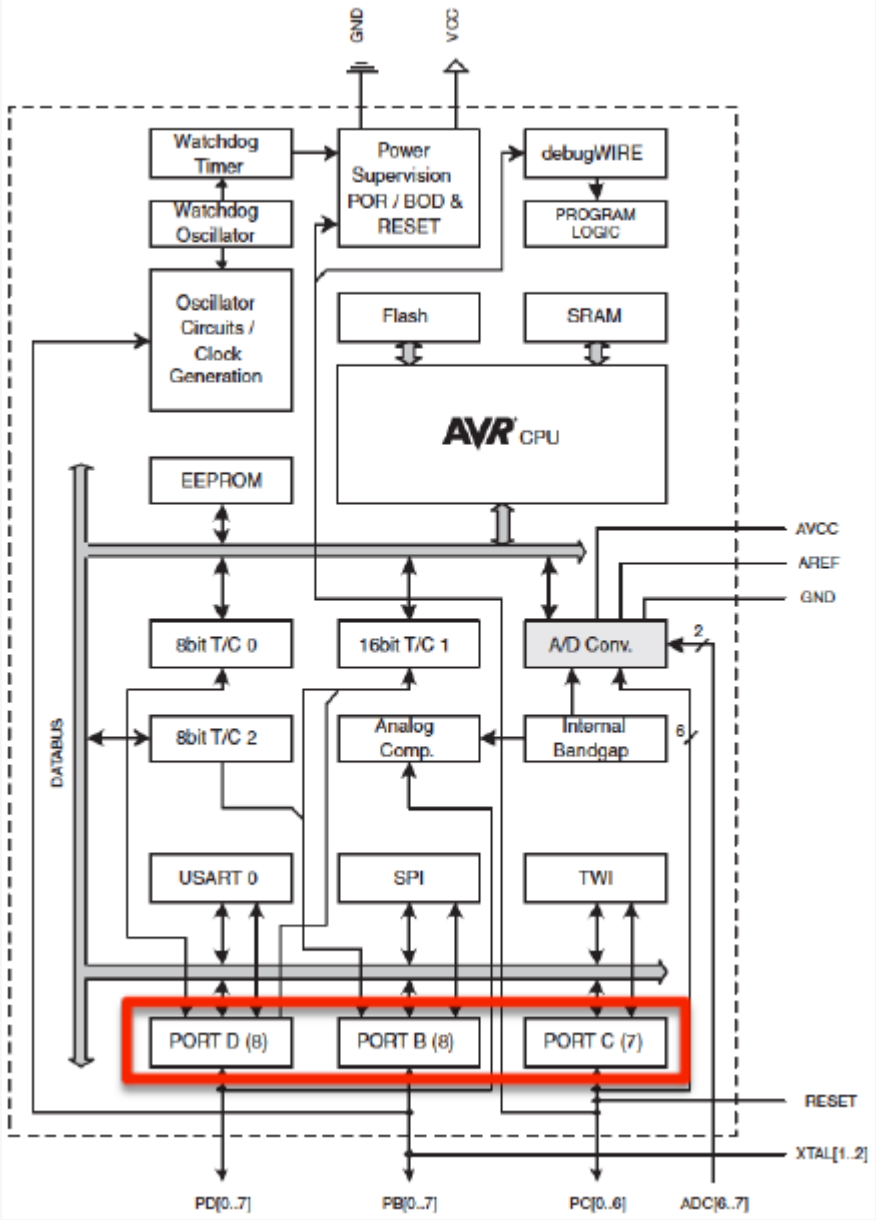
PORTx – Pin output value Writing this register sets value of pin



Arhitektura ATMEGA 328p

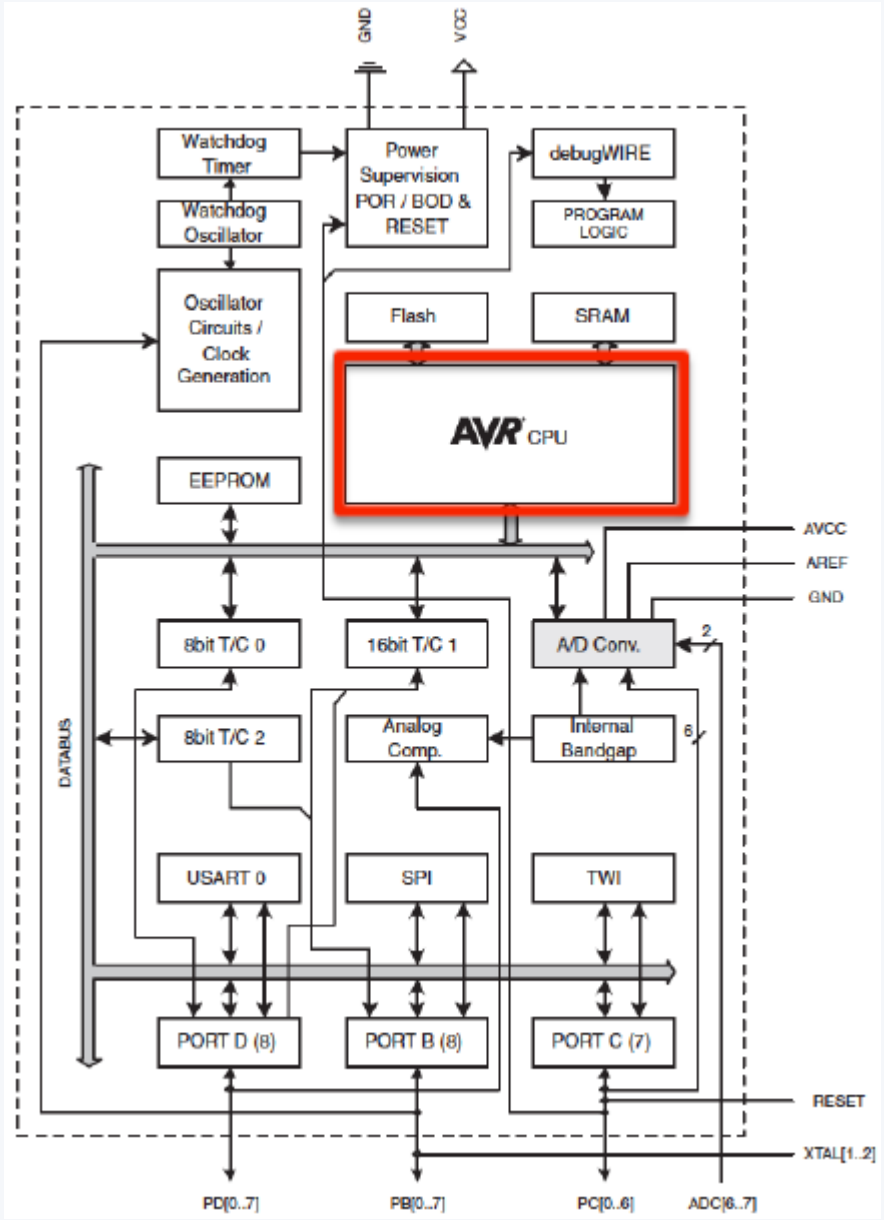


Note: 1. WRx, WPx, WDx, RRx, RPx, and RDx are common to all pins within the same port. $clk_{I/O}$, SLEEP, and PUD are common to all ports.



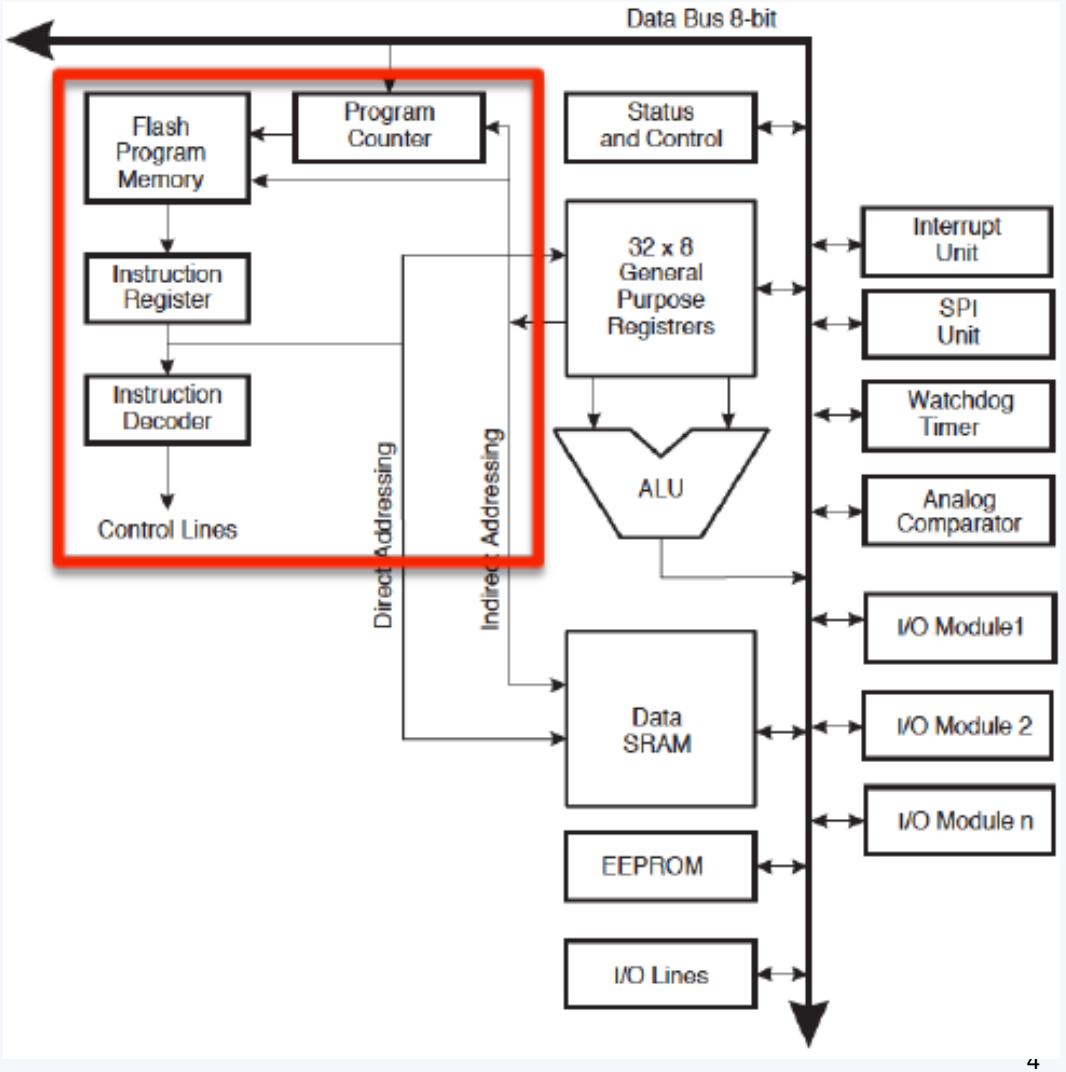
Arhitektura ATMEGA 328p

- PROCESOR - CPU



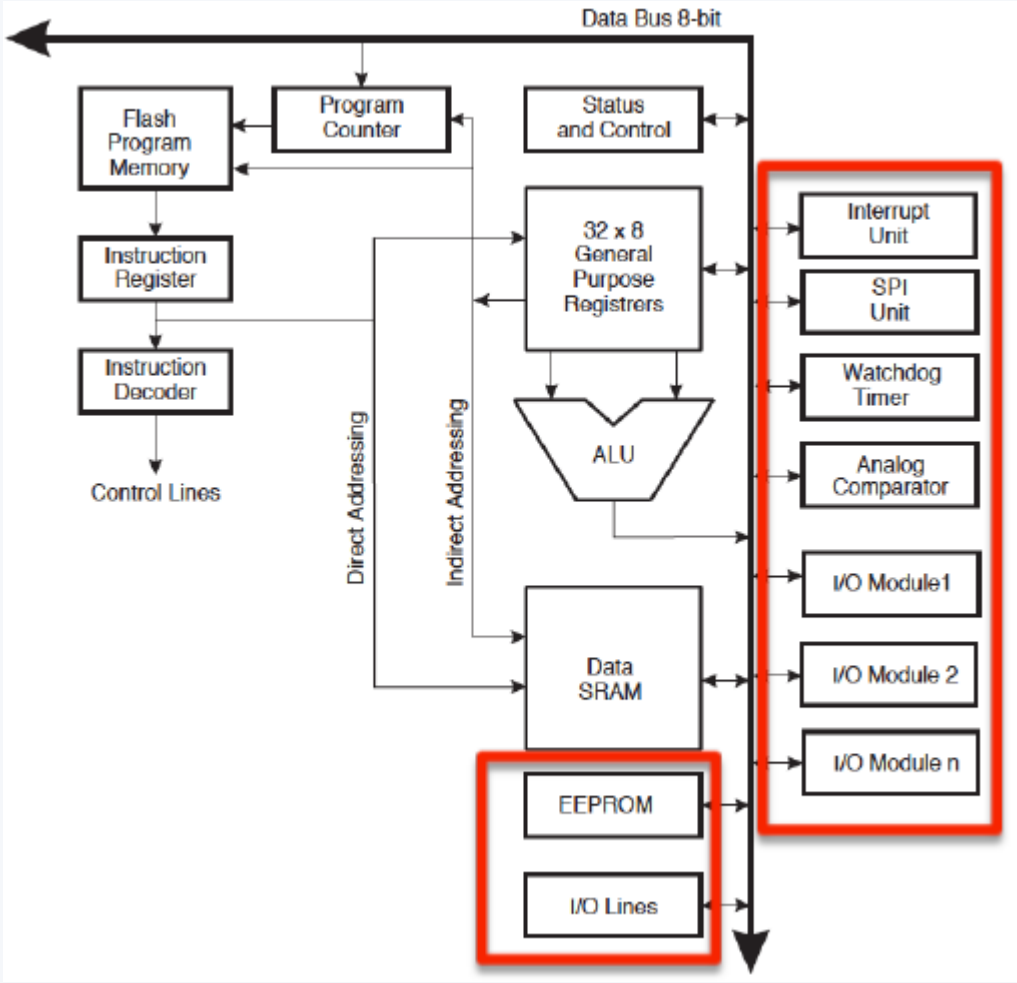
Arhitektura ATMEGA 328p - Procesor

- Dio koji pribavlja i dekodira instrukciju



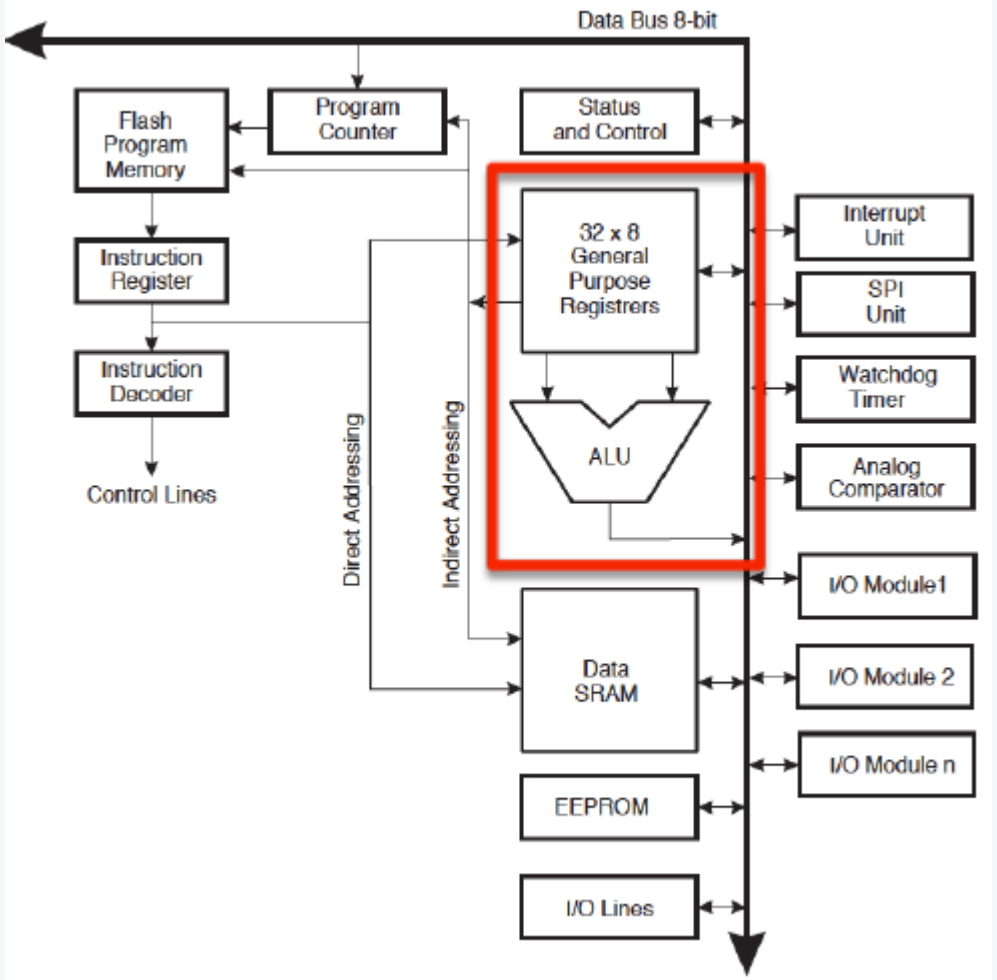
Arhitektura ATMEGA 328p - Procesor

- I/O moduli
- Moduli posebnih funkcija



Arhitektura ATMEGA 328p - Procesor

ALU + registri opće namjene



Arhitektura ATMEGA 328p - Procesor

Registri opće namjene

- 32 registra
- 8 bitna
- X, Y, Z – 16 bitni registri

Figure 6-2. AVR CPU General Purpose Working Registers

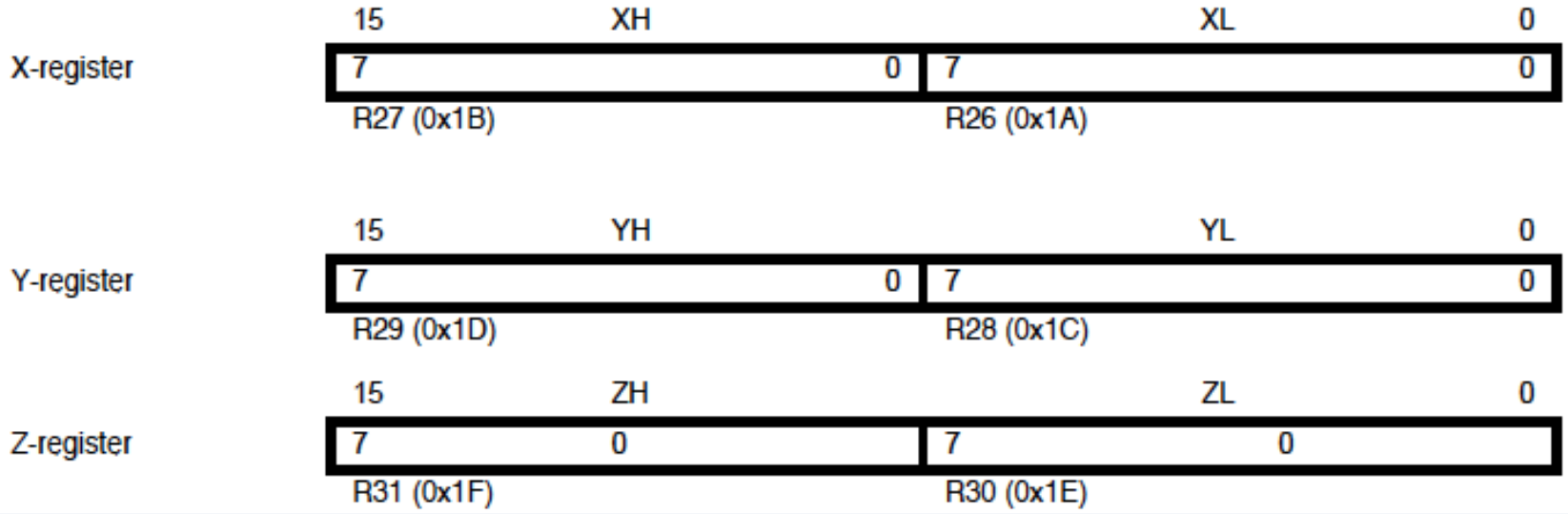
	7	0	Addr.	
General Purpose Working Registers	R0		0x00	
	R1		0x01	
	R2		0x02	
	...			
	R13		0x0D	
	R14		0x0E	
	R15		0x0F	
	R16		0x10	
	R17		0x11	
	...			
	R26		0x1A	X-register Low Byte
	R27		0x1B	X-register High Byte
	R28		0x1C	Y-register Low Byte
	R29		0x1D	Y-register High Byte
	R30		0x1E	Z-register Low Byte
	R31		0x1F	Z-register High Byte

Arhitektura ATMEGA 328p - Procesor

X, Y, Z

- 16 bitni registri
- Registri od R26 do R31
- nalaze se na adresama od 1B do 1F

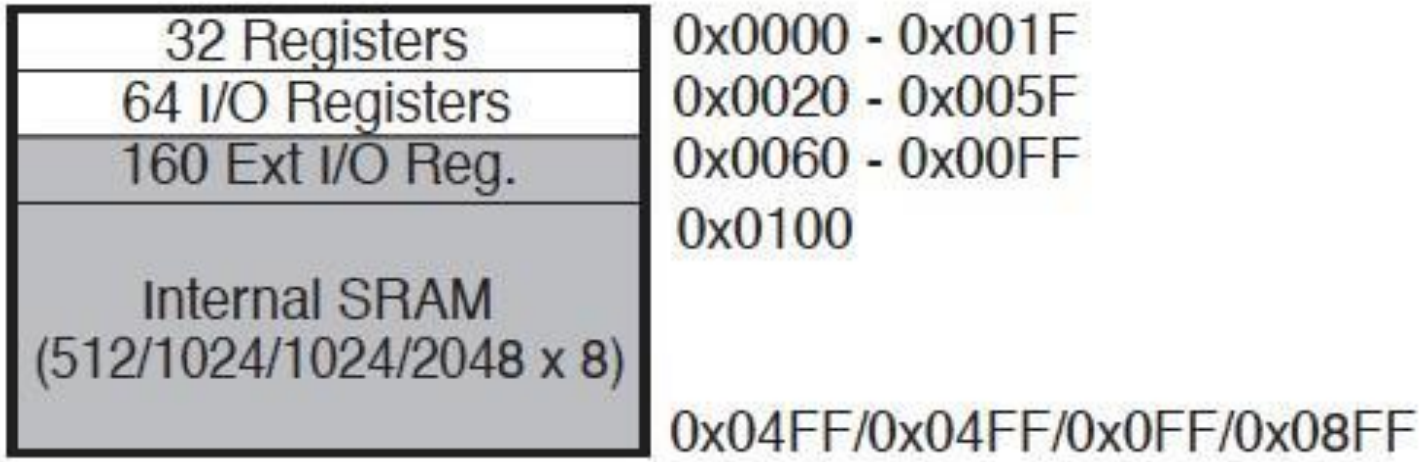
Figure 6-3. The X-, Y-, and Z-registers



Adresiranje

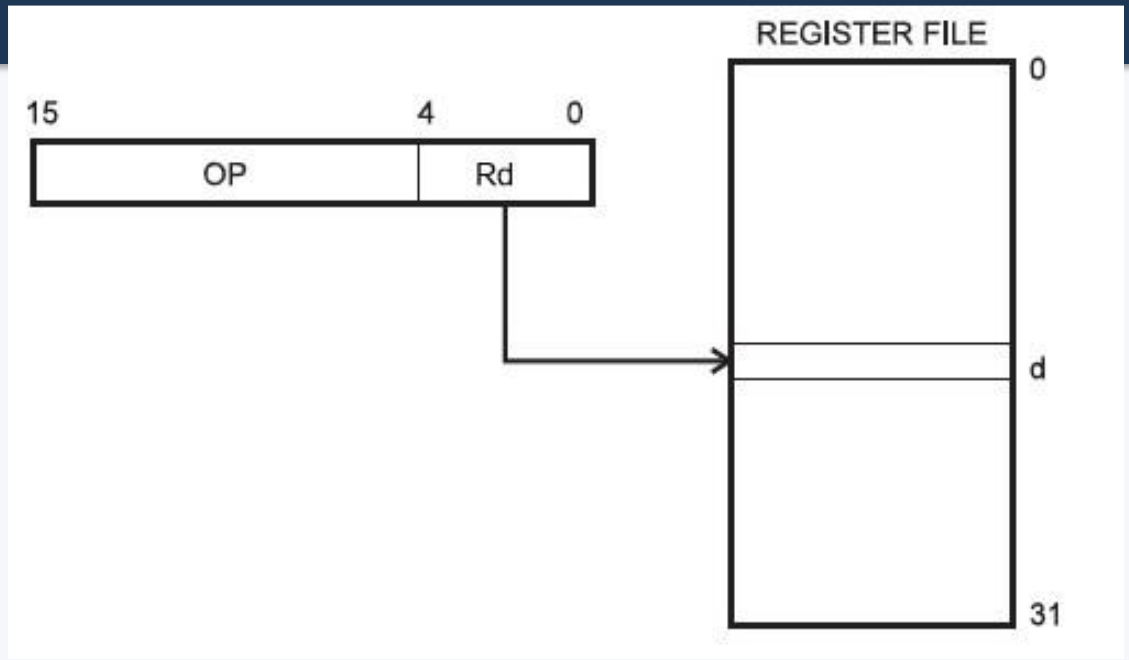
AVR memorija

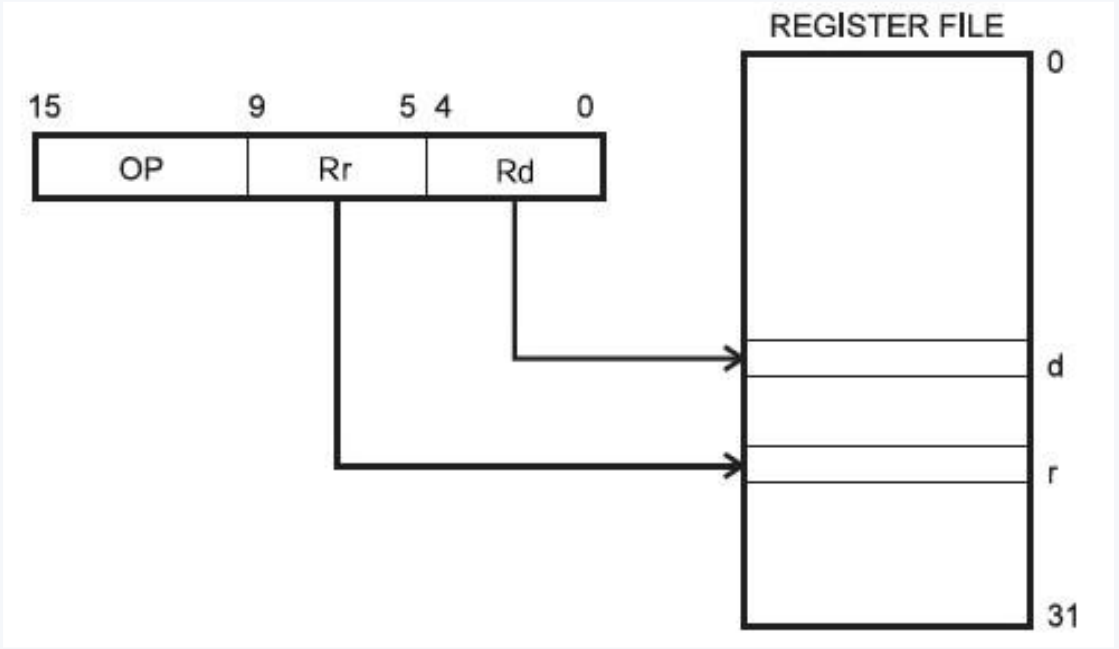
- Program memory – Flash
- Data memory - SRAM



Adresiranje

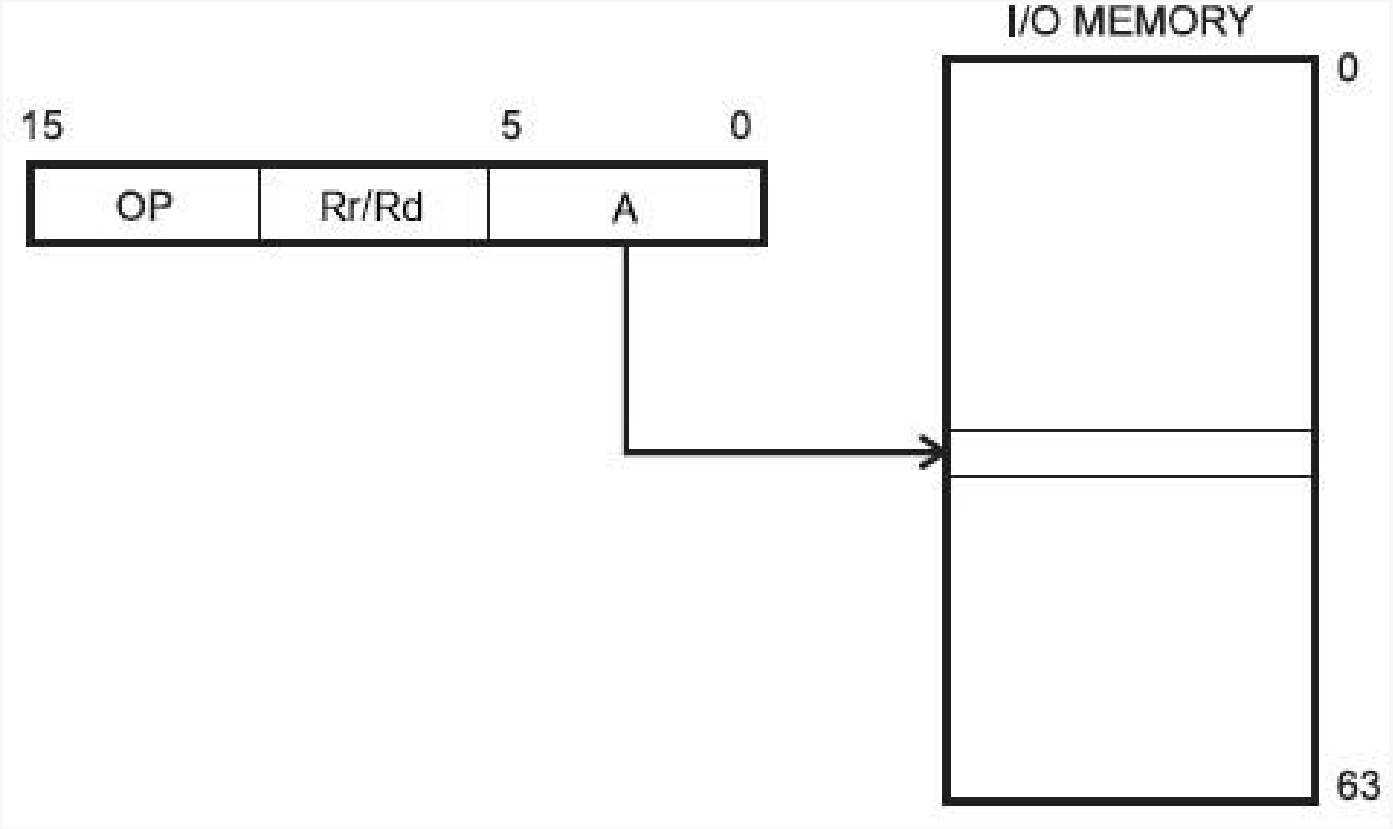
Direktno adresiranje





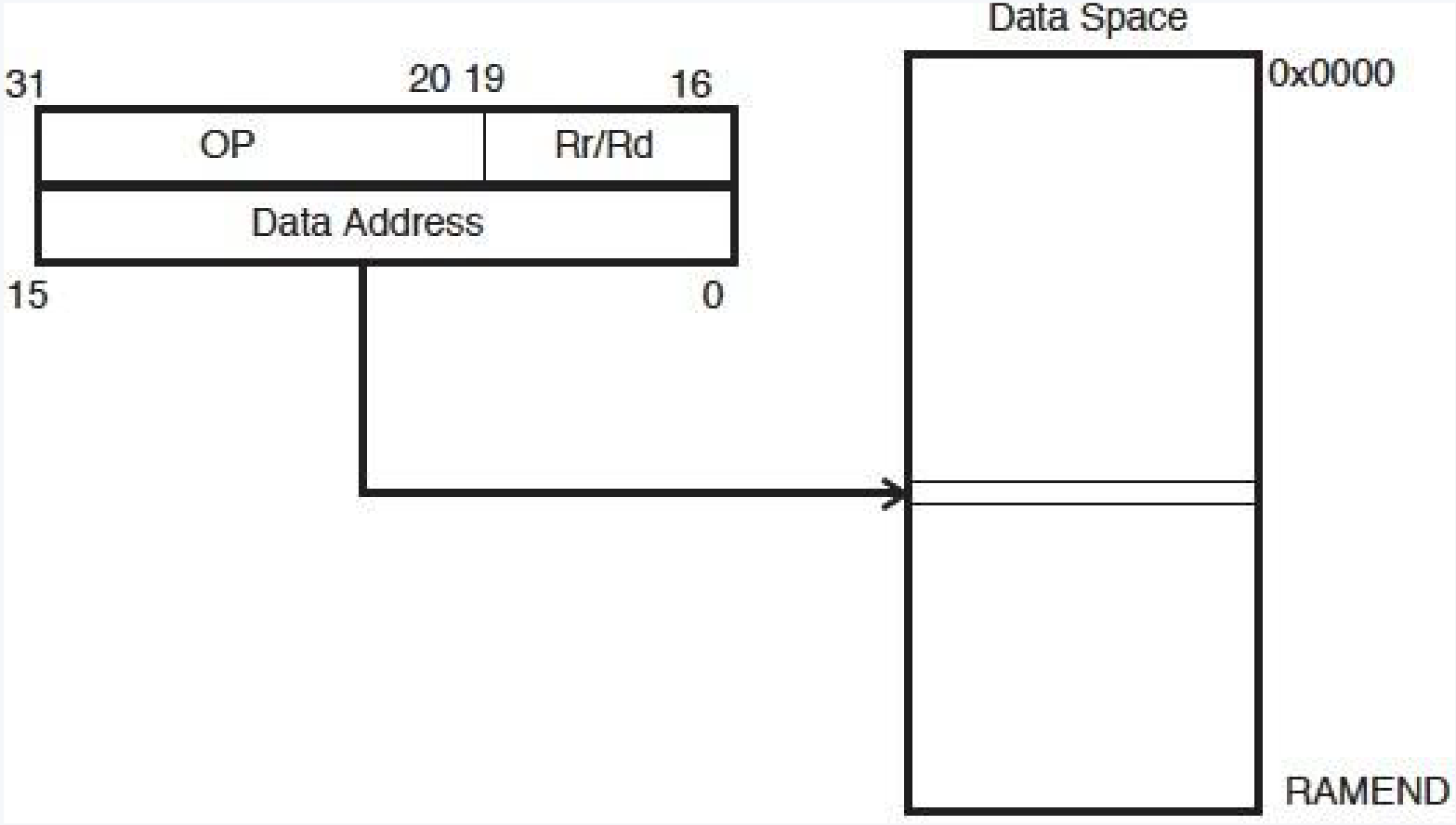
Adresiranje

Direktno I/O adresiranje



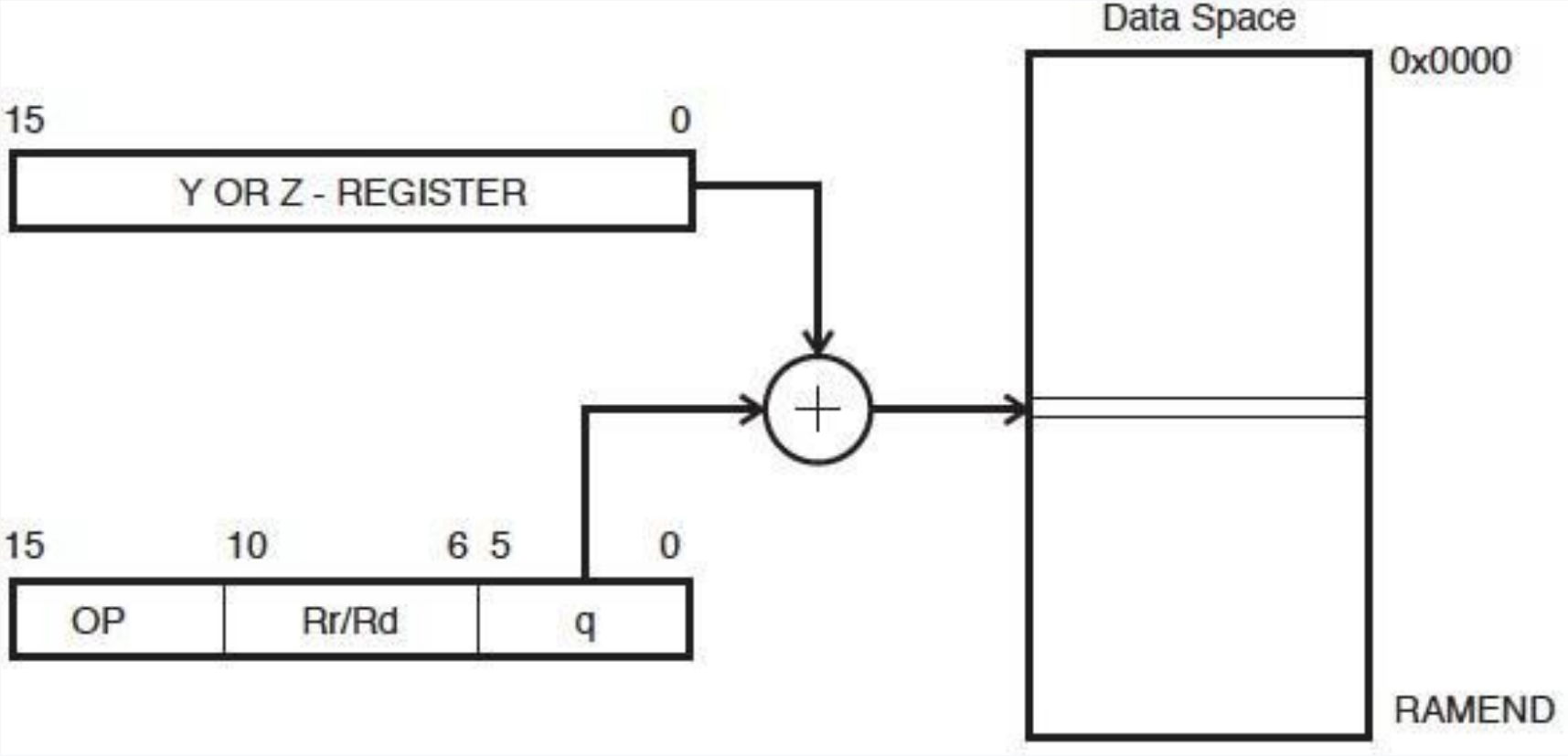
Adresiranje

Direktno adresiranje cijele memorije



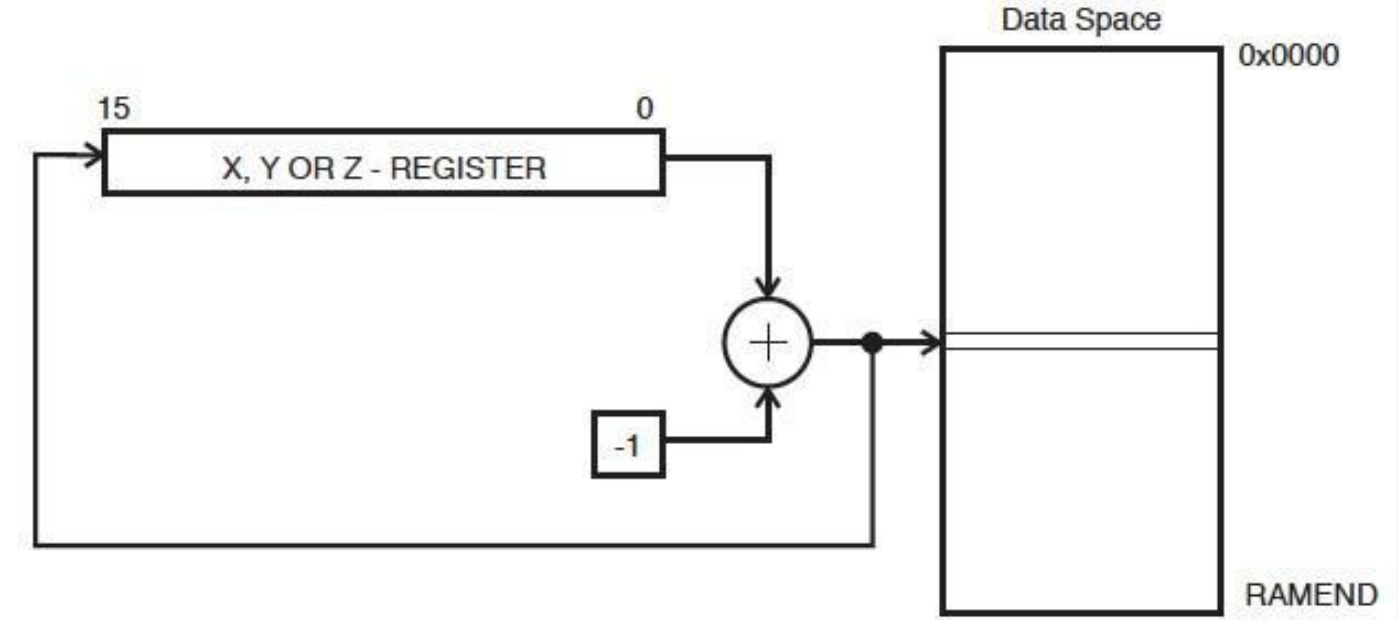
Adresiranje

Direktno adresiranje cijele memorije s pomakom



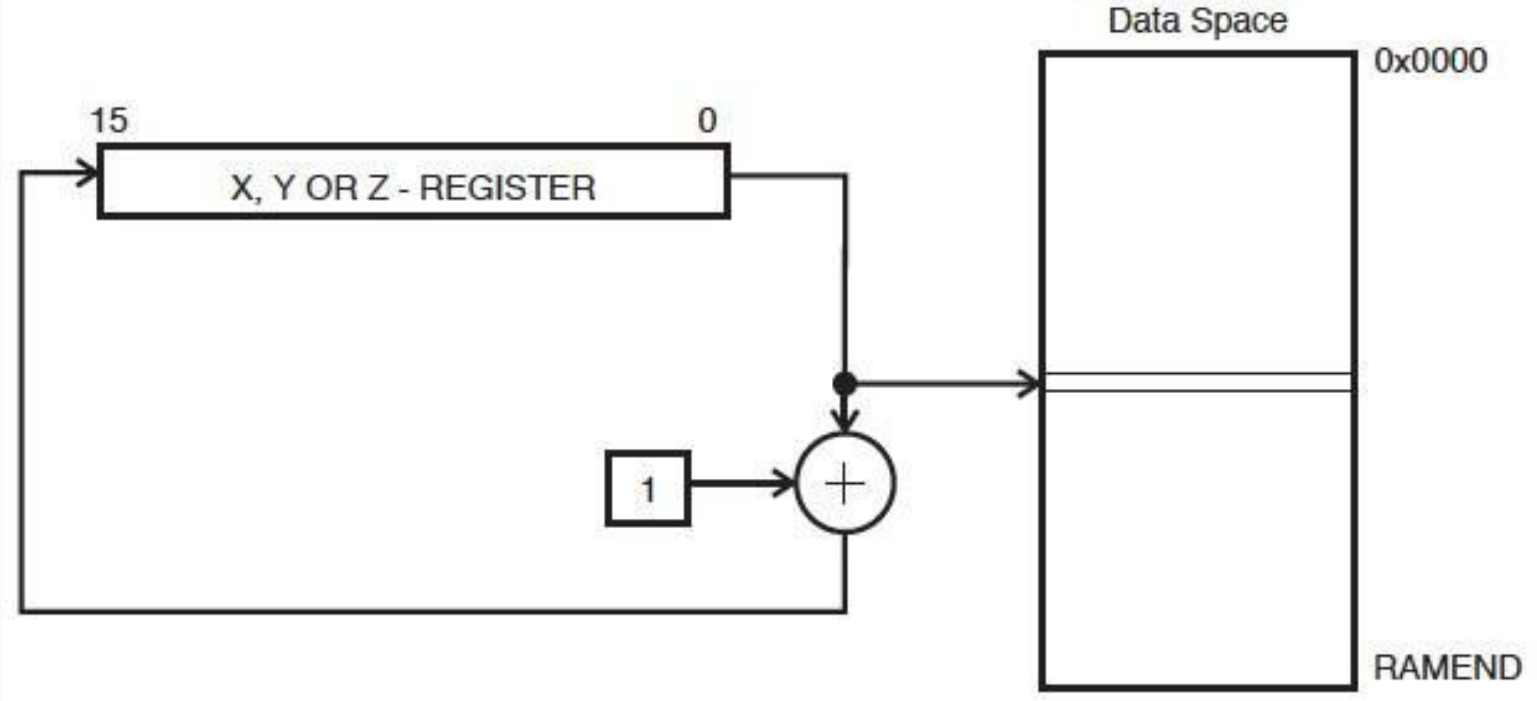
Adresiranje

indirektno adresiranje podatka u memoriji s prethodnim smanjenjem.



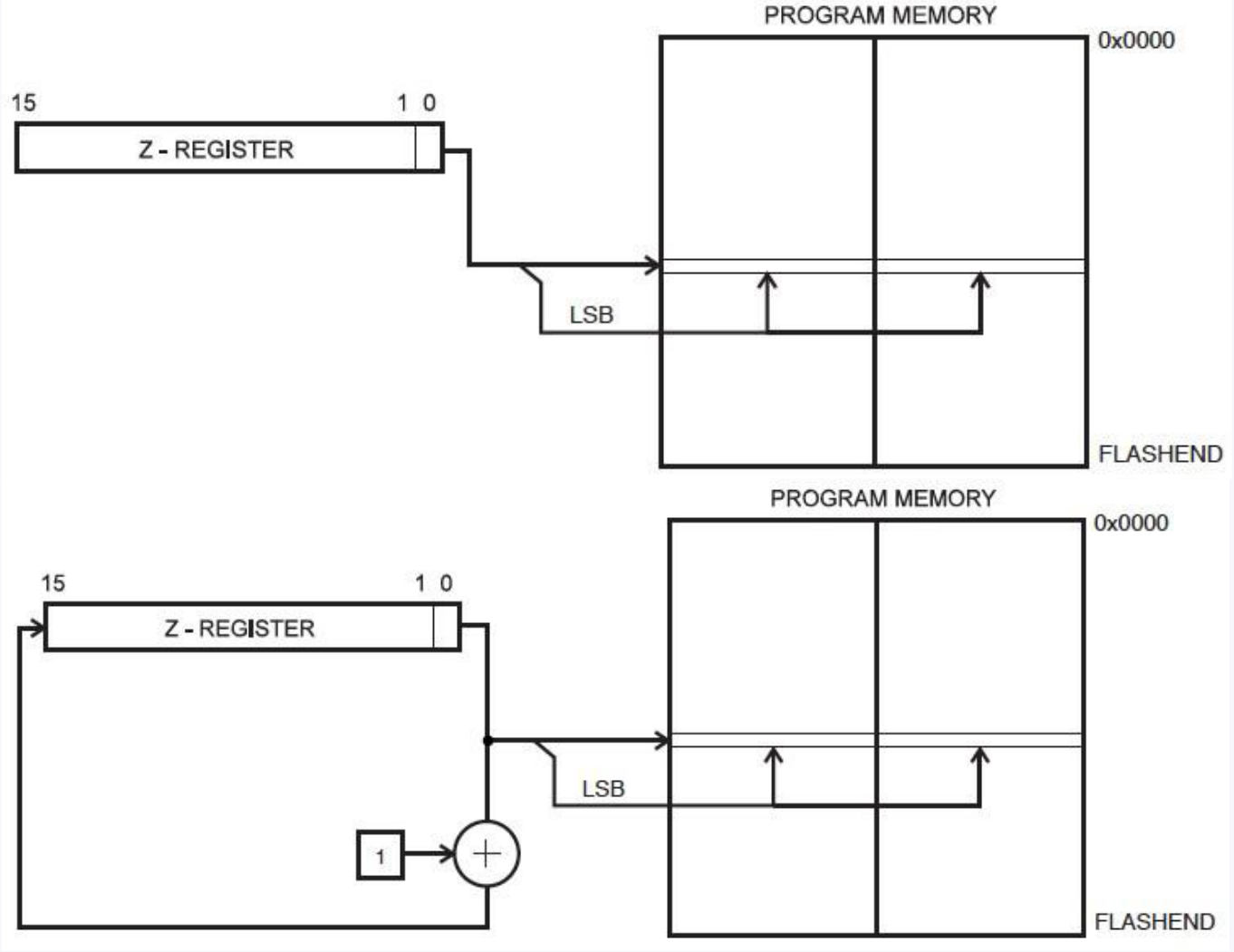
Adresiranje

indirektno adresiranje podatka u memoriji s naknadnim povećanjem.



Adresiranje

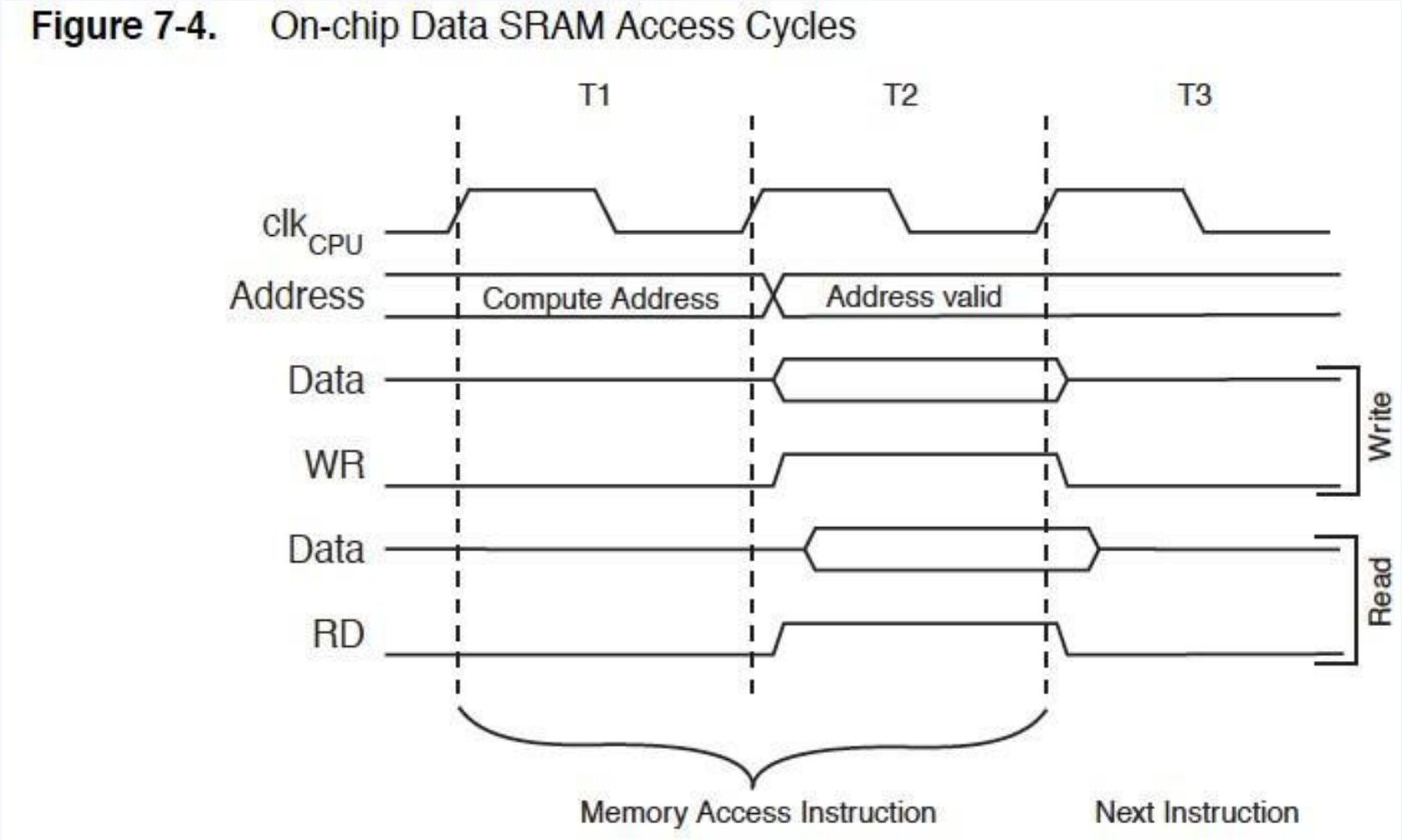
Adresiranje programske memorije



Adresiranje

Adresiranje programske memorije – vremenski pristup

Figure 7-4. On-chip Data SRAM Access Cycles



STACK

Poseban registar smješten u I/O prostoru, na adresama 0x3E i 0x3D.

- lokacije za gornji i donji bajt registra SP u prostoru I/O registara.
- dovoljno bitova da može adresirati cijeli podatkovni memorijski prostor.
- inicijalizira se na vrijednost RAMEND, odnosno na adresu posljednje lokacije RAM memorije

Table 6-1. Stack Pointer instructions

Instruction	Stack pointer	Description
PUSH	Decremented by 1	Data is pushed onto the stack
CALL ICALL RCALL	Decremented by 2	Return address is pushed onto the stack with a subroutine call or interrupt
POP	Incremented by 1	Data is popped from the stack
RET RETI	Incremented by 2	Return address is popped from the stack with return from subroutine or return from interrupt

Program status register (PRS)

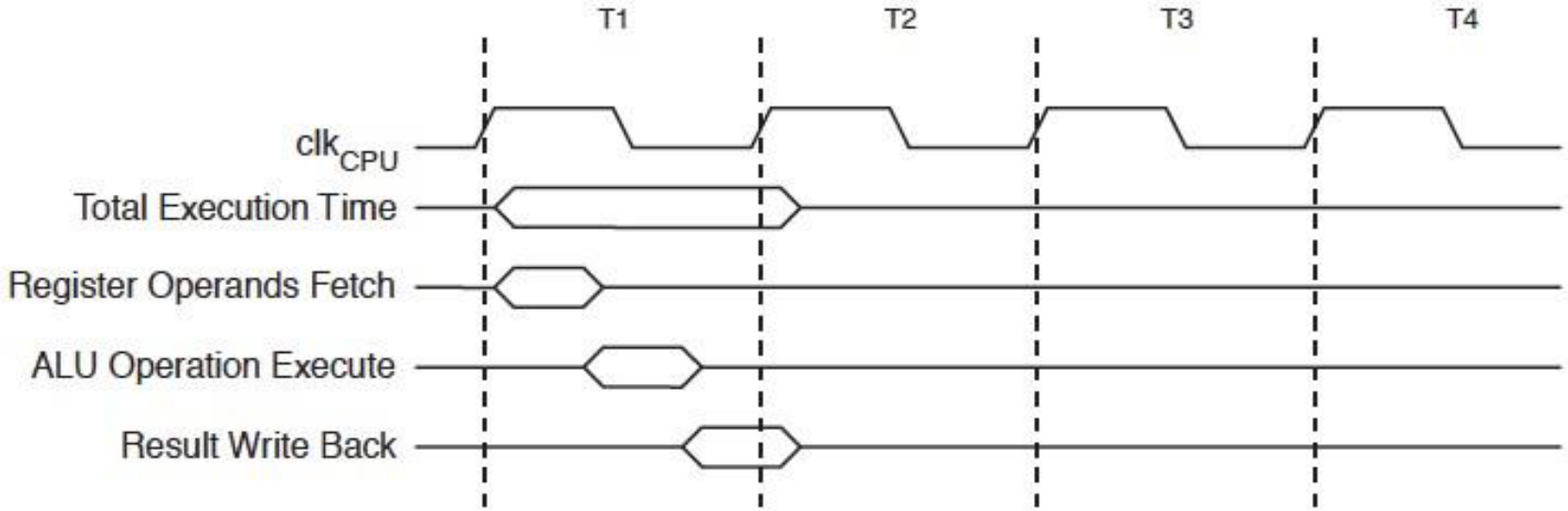
Zastavice
I – Global interrupt enable
T – Flag bit
H – Half carry (BCD arithmetic)
S – Sign
V – Overflow
N – Negative
Z – Zero
C – Carry

Bit	7	6	5	4	3	2	1	0	
0x3F (0x5F)	I	T	H	S	V	N	Z	C	SREG
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

Izvođenje instrukcije

- Instrukcije ALU-a se izvršavaju u jednom ciklusu
- relativno jednostavan izračun vremena izvođenja programa.
- NO cache misses).
- Jedan takt procesora po instrukciji (1 takt/instrukcija).

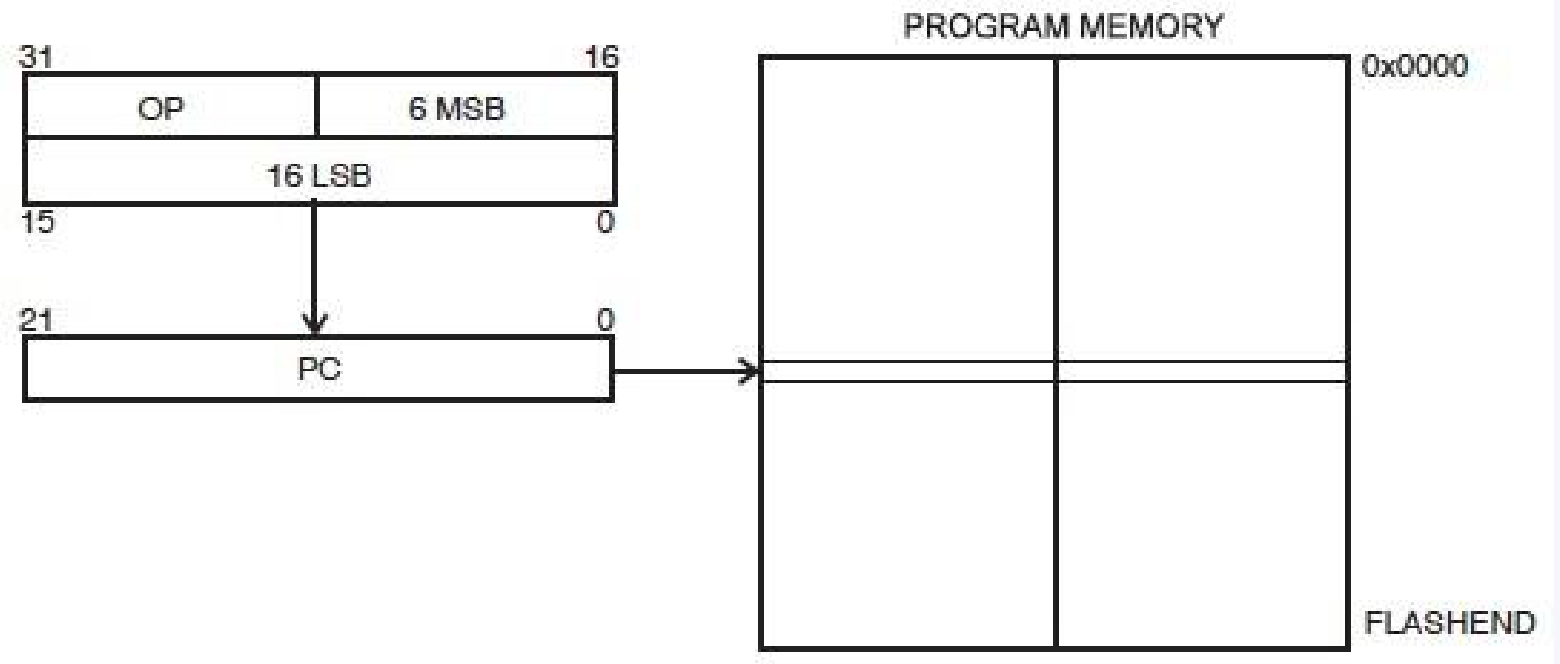
Figure 6-5. Single Cycle ALU Operation



Adresiranje programske memorije

Direktno adresiranje pomoću

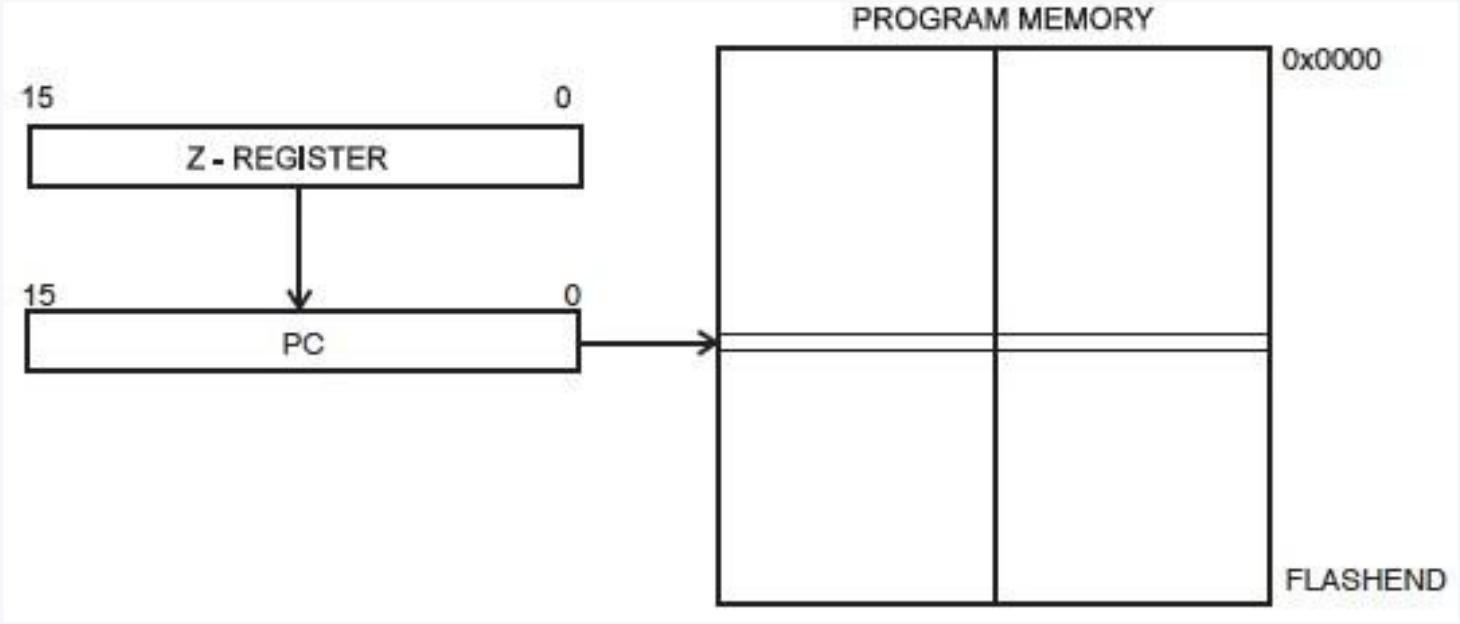
- JMP
- CALL



Adresiranje programske memorije

INdirektno adresiranje pomoću

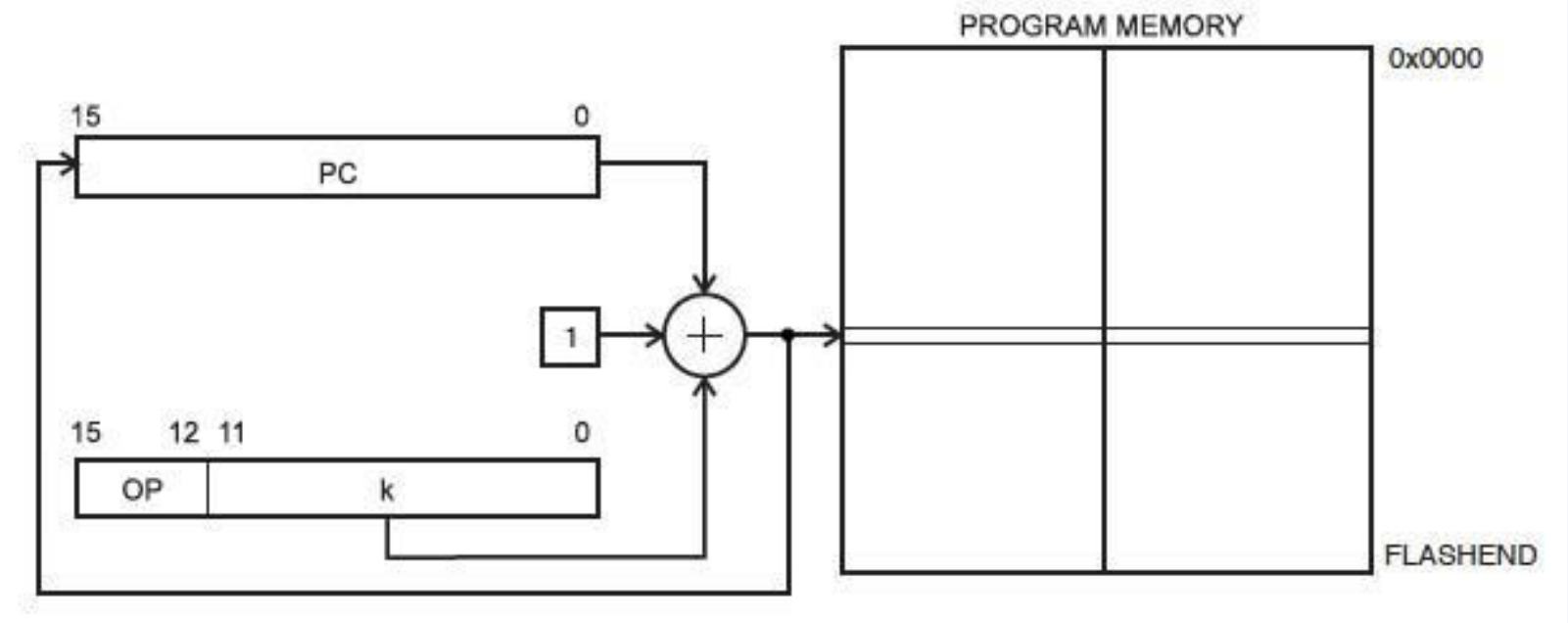
- IJMP
- ICALL



Adresiranje programske memorije

Relativno adresiranje pomoću

- RJMP
- RCALL



Hvala na pažnji!